MACROEVOLUTION

Module map

- Module 14: Macroevolution
- Connected lab: lab-15_macroevolution.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 14
Macroevolution

1. Species concepts
2. Reproductive isolation
3. Prezygotic barriers
4. Postzygotic barriers
5. Speciation patterns and phylogeny

Lab evidence
lab-15_macroevolution.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

MACROEVOLUTION

Learning objectives

- State the biological species concept and its limits.
- Explain reproductive isolation as reduced gene flow.
- Classify examples as prezygotic or postzygotic barriers.
- Compare allopatric and sympatric speciation.
- Interpret speciation as population-level divergence over time.

OBJECTIVE 1
State the biological species concept and its limits.

OBJECTIVE 2
Explain reproductive isolation as reduced gene flow.

OBJECTIVE 3
Classify examples as prezygotic or postzygotic barriers.

OBJECTIVE 4
Compare allopatric and sympatric speciation.

OBJECTIVE 5
Interpret speciation as population-level divergence over time.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

MACROEVOLUTION

Topic sequence

- Species concepts: The biological species concept uses natural interbreeding, viable fertile offspring, and isolation. It is difficult to apply to fossils or asexual organisms.
- Reproductive isolation: Reproductive isolation reduces gene flow. Appearance or separation alone does not establish species boundaries; some distinct species hybridize.
- Prezygotic barriers: Prezygotic barriers act before fertilization: habitat, timing, behavior, anatomy, or gamete incompatibility can prevent a zygote.
- Postzygotic barriers: Postzygotic barriers reduce hybrid development, survival, or reproduction. Classify the stage using the evidence provided.
- Speciation patterns and phylogeny: Allopatric speciation begins with separation; sympatric speciation does not. Polyploid lineages must reproduce. Phylogenies show branching ancestry.

01

Species concepts

The biological species concept uses natural interbreeding, viable fertile offspring, and isolation. It is difficult to apply to fossils or asexual organisms.

02

Reproductive isolation

Reproductive isolation reduces gene flow. Appearance or separation alone does not establish species boundaries; some distinct species hybridize.

03

Prezygotic barriers

Prezygotic barriers act before fertilization: habitat, timing, behavior, anatomy, or gamete incompatibility can prevent a zygote.

04

Postzygotic barriers

Postzygotic barriers reduce hybrid development, survival, or reproduction. Classify the stage using the evidence provided.

05

Speciation patterns and phylogeny

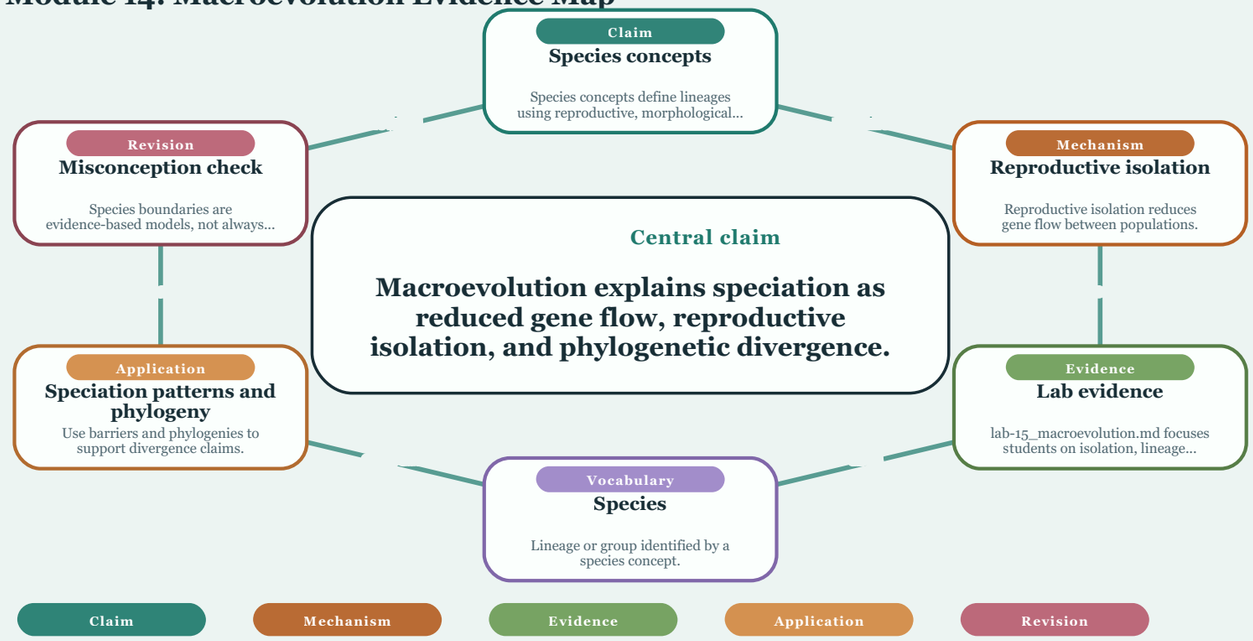
Allopatric speciation begins with separation; sympatric speciation does not. Polyploid lineages must reproduce. Phylogenies show branching ancestry.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

MACROEVOLUTION

Module 14: Macroevolution Evidence Map

- Module focus: Macroevolution
- Central claim: Macroevolution explains speciation as reduced gene flow, reproductive isolation, and phylogenetic divergence.
- Key nodes to connect: Species concepts, Reproductive isolation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-15_macroevolution.md supplies an evidence surface for the map.



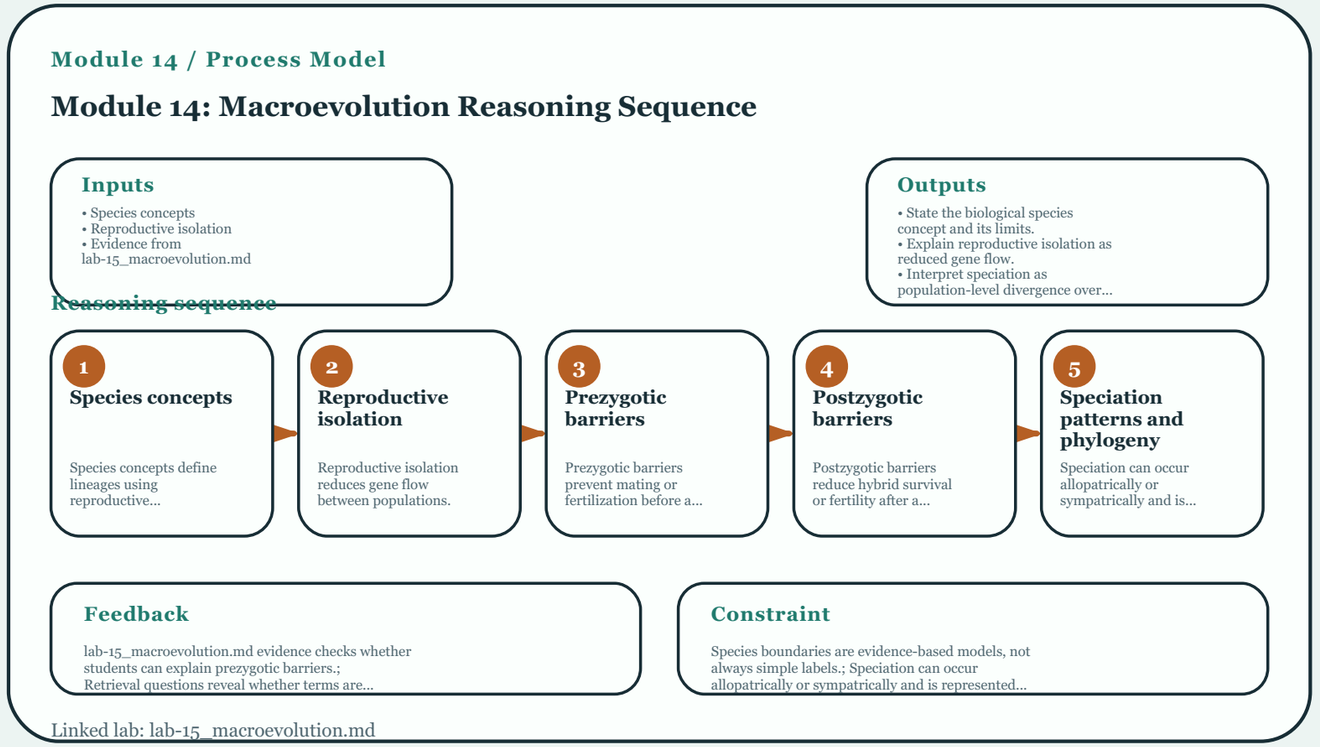
Linked lab: lab-15_macroevolution.md / Generated from explicit module.toml visual schema

Teaching note: Ask students to explain one edge in their own words before moving on.

MACROEVOLUTION

Module 14: Macroevolution Reasoning Sequence

- Module focus: Macroevolution
- Trace the model: Species concepts -> Reproductive isolation -> Prezygotic barriers -> Postzygotic barriers
- Inputs become outputs: Species concepts, Reproductive isolation -> State the biological species concept and its limits., Explain reproductive isolation as reduced gene flow.
- Feedback check: lab-15_macroevolution.md evidence checks whether students can explain prezygotic barriers.
- Lab connection: lab-15_macroevolution.md gives students a process to observe or test.



Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

MACROEVOLUTION

Terms as evidence handles

- Species: Lineage or group identified by a species concept.
- Biological species concept: Species defined by interbreeding and reproductive isolation.
- Reproductive isolation: Barriers that reduce gene flow between groups.
- Prezygotic barrier: Barrier before fertilization.
- Postzygotic barrier: Barrier after fertilization.
- Allopatric speciation: Speciation with geographic separation.

SPECIES
Lineage or group identified by a species concept.

BIOLOGICAL SPECIES CONCEPT
Species defined by interbreeding and reproductive isolation.

REPRODUCTIVE ISOLATION
Barriers that reduce gene flow between groups.

PREZYGOTIC BARRIER
Barrier before fertilization.

POSTZYGOTIC BARRIER
Barrier after fertilization.

ALLOPATRIC SPECIATION
Speciation with geographic separation.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

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Lab connection

- Lab file: lab-15_macroevolution.md
- Lab evidence should test or illustrate: Prezygotic barriers
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Species concepts

EVIDENCE
lab-15_macroevolution.md

CLAIM
State the biological species concept and its limits.

Teaching note: Make the lab connection explicit before students encounter the activity.

MACROEVOLUTION

Contrast check

- Do not stop at: The biological species concept uses natural interbreeding, viable fertile offspring, and isolation. It is difficult to apply to fossils or asexual organisms.
- Stronger explanation: Allopatric speciation begins with separation; sympatric speciation does not. Polyploid lineages must reproduce. Phylogenies show branching ancestry.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

The biological species concept uses natural interbreeding, viable fertile offspring, and isolation. It is difficult to apply to fossils or asexual organisms.

DEEPER

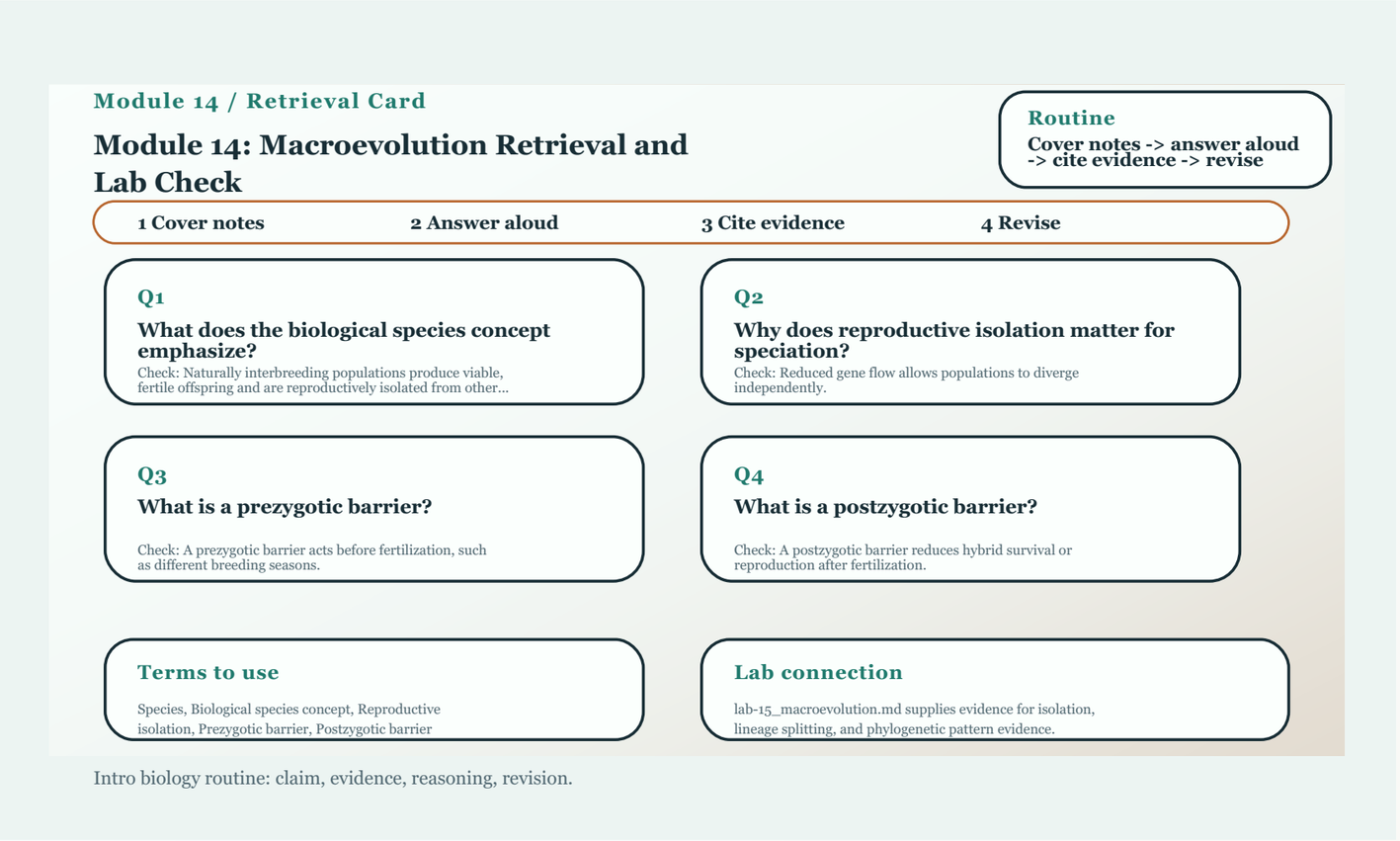
Allopatric speciation begins with separation; sympatric speciation does not. Polyploid lineages must reproduce. Phylogenies show branching ancestry.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

MACROEVOLUTION

Module 14: Macroevolution Retrieval and Lab Check

- Module focus: Macroevolution
- Retrieval prompt: What does the biological species concept emphasize?
- Answer check: Naturally interbreeding populations produce viable, fertile offspring and are reproductively isolated from other groups.
- Required terms: Species, Biological species concept, Reproductive isolation
- Lab connection: lab-15_macroevolution.md; lab-15_macroevolution.md supplies evidence for isolation, lineage splitting, and phylogenetic pattern evidence.



Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

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Practice quiz bridge

- 1. Under the biological species concept, two groups count as different species when they:
- 2. Two related species breed at different times of year and so never mate. This is an example of a:
- 3. A mountain range splits one population into two groups that slowly diverge into separate species. This is:
- 4. A mule, the offspring of a horse and a donkey, is healthy but sterile. This illustrates a:

QUESTION 1

Key: C

Under the biological species concept, two groups count as different species when they:

QUESTION 2

Key: A

Two related species breed at different times of year and so never mate. This is an example of a:

QUESTION 3

Key: D

A mountain range splits one population into two groups that slowly diverge into separate species. This is:

QUESTION 4

Key: B

A mule, the offspring of a horse and a donkey, is healthy but sterile. This illustrates a:

Teaching note: Use the quiz as formative feedback. Answer key: 1=C, 2=A, 3=D, 4=B

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Synthesis and exit ticket

- Exit claim: explain species concepts using evidence from lab-15_macroevolution.md.
- Must include: Species, Biological species concept, Reproductive isolation.
- Revision prompt: what evidence would change your explanation?

CLAIM
Species concepts

EVIDENCE
lab-15_macroevolution.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.